

Horizon Hysterical Response

Analytic agreement with Hawking radiation and Tolman temperature

Andrew Bruce Boyd

Working mathematical formulation — September 2026

Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

Hysterical Response Theory (HRT) isolates the residual response of an already-existing interaction under open-system dynamics, $F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S})$ [1]. Applied at a nonextremal stationary horizon through the membrane / open-system embedding of the exterior [5, 4, 3], that package reproduces analytically the standard semiclassical predictions of Hawking radiation [2] and Tolman local temperature [6, 7]. The Euclidean period fixed by horizon regularity supplies $T_H = \kappa/(2\pi)$; quantum fluctuation–dissipation and detailed balance yield the Planck occupation; exterior greybodies filter the asymptotic flux; static equilibrium continues the same KMS scale to $T_{\text{loc}}(r) = T_H/\sqrt{1 - 2M/r}$ for $r > 2M$. We record the structural dictionary that makes this an effective-method equivalence, not merely a similarity of shape, and interpret why Hawking (fluctuation branch at a causal dissipative boundary) and Tolman (static equilibrium profile of that thermal response) each touched the Hysterical Response without naming it. Geometry-free derivation of the Euclidean period from a Hysterical action alone is not claimed; Hawking likewise takes that period from horizon regularity.

Contents

1	Scope and status	2
2	Horizon regularity fixes the response temperature	2
3	Causal horizon memory as Hysterical response	3
4	Detailed balance produces the Hawking factor	3
5	Tolman exterior temperature	4
6	Why Hawking and Tolman touched the Hysterical Response	5
7	Backreaction as feedback	5
8	Logical status of the results	6
9	Conclusion	6

1 Scope and status

The response-theory paper defines the Hysterical Response as the residual component of an already-existing interaction under exponentially stable reduced dynamics [1]:

$$F_H := -\text{Tr}(F \mathcal{L}^{-1} \mathcal{S}). \quad (1)$$

No new force carrier is introduced. The present paper asks what that package predicts when the interaction channel is a quantum field on a Schwarzschild exterior with the horizon represented as a causal, dissipative boundary.

The claim is

$$\boxed{\begin{array}{l} \text{HRT horizon response} + \text{membrane/open-system embedding} + \text{horizon regularity} \\ \implies \text{Hawking asymptotics and Tolman static temperature (analytic agreement).} \end{array}} \quad (2)$$

Four layers must be kept separate.

Prior HRT	Resolvent response (1); open-system FDR structure (cited from [1]).
Literature hypothesis	Membrane paradigm; influence-functional noise/dissipation; Euclidean regularity $\Rightarrow$ KMS period; Tolman's static equilibrium law.
Modeling assumption	Horizon Langevin packaging of the exterior response.
Theorem (this paper)	Algebraic occupation, temperature, Tolman redshift, and mass-loss identities under those inputs.

2 Horizon regularity fixes the response temperature

Assumption 2.1 (Nonextremal stationary horizon). The spacetime admits a nonextremal Killing horizon with surface gravity $\kappa > 0$. Near the horizon the metric reduces locally to Rindler form

$$ds^2 \simeq -(\kappa\rho)^2 dt^2 + d\rho^2 + r_H^2 d\Omega^2. \quad (3)$$

Literature Hypothesis 2.2 (Euclidean regularity). Under $t = -i\tau$, regularity at $\rho = 0$ requires the identification $\tau \sim \tau + 2\pi/\kappa$ [2, 8]. A regular horizon state is therefore KMS at inverse temperature $\beta_H = 2\pi/\kappa$ (units $\hbar = c = k_B = 1$).

Theorem 2.3 (Hawking temperature from the Euclidean period). *If $\beta_H = 2\pi/\kappa$ with $\kappa \neq 0$, then*

$$T_H := \beta_H^{-1} = \frac{\kappa}{2\pi}. \quad (4)$$

For Schwarzschild, $\kappa = 1/(4M)$ yields $T_H = 1/(8\pi M)$.

Proof. Immediate from $\beta_H = 2\pi/\kappa$. □

Remark 2.4 (Shared geometric input). Equation (4) uses horizon geometry exactly as in Hawking's derivation. We do not claim to derive β_H from a Hysterical action with the metric deleted.

3 Causal horizon memory as Hysterical response

Literature Hypothesis 3.1 (Membrane / open-system embedding). The exterior influence of the horizon is represented by a stretched, dissipative membrane [5, 4]. Influence-functional treatments of black-hole backreaction likewise yield nonlocal dissipation and stochastic noise together, with the noise linked to Bogoliubov particle production [3].

Modeling Assumption 3.2 (Horizon Langevin response). For a response coordinate h (e.g. an electromagnetic multipole on the stretched horizon),

$$\dot{h}(t) + \int_{-\infty}^t \Gamma(t-t')h(t') dt' = \sigma A_{\text{in}}(t) + \xi(t), \quad (5)$$

with causal memory kernel Γ , incoming drive A_{in} , and intrinsic noise ξ . In frequency space,

$$h(\omega) = \chi_R(\omega)[\sigma A_{\text{in}}(\omega) + \xi(\omega)], \quad A_{\text{out}}(\omega) = \alpha(\omega)h(\omega). \quad (6)$$

Proposition 3.3 (Hysterical identification). *Equation (5) is Hysterical in the sense of [1]: the present response depends on the previous trajectory through a retarded kernel. The susceptibility χ_R is the open-system linear response of the horizon channel; ξ is the fluctuation partner of dissipation, not an externally inserted Hawking flux.*

	HRT object	Horizon realization
Definition 3.4 (Structural dictionary).	Retarded memory / χ_R	Membrane susceptibility (6)
	Dissipation $\text{Im } \Gamma_R$	Horizon absorption
	Noise ξ / S_ξ	Spontaneous fluctuation branch
	Spatial filter	Exterior greybody $\mathcal{G}$
	Feedback variable	Black-hole mass M (backreaction)

4 Detailed balance produces the Hawking factor

Literature Hypothesis 4.1 (Quantum FDR at the horizon period). With β_H from Literature Hypothesis 2.2, the unsymmetrized noise obeys the KMS detailed-balance relation

$$S_\xi(-\omega) = e^{-\beta_H \omega} S_\xi(+\omega), \quad (7)$$

equivalently a quantum fluctuation–dissipation relation tying the symmetrized noise to $\text{Im } \Gamma_R$ [3, 8].

Lemma 4.2 (Detailed-balance occupation). *If $r \neq 1$, $p_+ \neq 0$, and $p_- = r p_+$, then*

$$\frac{p_-}{p_+ - p_-} = \frac{r}{1 - r}.$$

Proof. Substitute $p_- = r p_+$ and clear the common factor. □

Lemma 4.3 (Exponential ratio is Bose). *If $e^x \neq 1$, then*

$$\frac{e^{-x}}{1 - e^{-x}} = \frac{1}{e^x - 1}.$$

Proof. Rewrite $e^{-x} = 1/e^x$ and clear denominators. □

Theorem 4.4 (Hawking occupation from HRT detailed balance). *Under Literature Hypotheses 2.2–4.1 and Model 3.2, set $r = e^{-\beta_H \omega}$. Then the detailed-balance occupation is*

$$n_\omega = \frac{1}{e^{\beta_H \omega} - 1}. \quad (8)$$

The asymptotic number flux takes the standard form

$$\frac{dN}{dt d\omega} = \frac{1}{2\pi} \sum_{\ell, m, p} \frac{\mathcal{G}_{\ell mp}(\omega)}{e^{\beta_H \omega} - 1}, \quad (9)$$

where $\mathcal{G}$ is the exterior greybody transmission factor.

Proof. Lemma 4.2 with $r = e^{-\beta_H \omega}$ and Lemma 4.3 with $x = \beta_H \omega$ give (8). Greybodies encode propagation through the exterior curvature potential and multiply the universal thermal denominator fixed by the horizon response [2]. $\square$

Theorem 4.5 (Effective-method agreement with Hawking). *Under the membrane/open-system embedding (Literature Hypothesis 3.1) and the horizon KMS/FDR package (Literature Hypotheses 2.2, 4.1), the HRT horizon-response calculation of Theorem 4.4 agrees analytically with Hawking’s asymptotic thermal prediction: same $T_H = \kappa/(2\pi)$, same Planck factor, and same greybody-filtered flux. At this effective-response abstraction the two methods are the same calculation, not merely similar shapes.*

Proof. Hawking’s Euclidean/Bogoliubov analysis supplies precisely the period β_H , the thermal detailed-balance ratio, and the exterior transmission problem that enter Theorem 4.4 [2, 3]. The HRT packaging identifies those objects with the noise and susceptibility of Definition 3.4. The algebraic output is identical. $\square$

Remark 4.6 (Mean versus covariance). At fixed mass the classical quasinormal poles remain stable. Hawking production is growth of quantum covariance, $\langle A_{\text{out}} \rangle = 0$ while $\langle A_{\text{out}}^\dagger A_{\text{out}} \rangle = n_\omega > 0$, not exponential growth of a classical mean field.

5 Tolman exterior temperature

Literature Hypothesis 5.1 (Tolman equilibrium law). In a static gravitational field, thermal equilibrium requires $T_{\text{loc}} \sqrt{-g_{tt}} = \text{const}$ [6, 7].

Assumption 5.2 (Static exterior observers). Restrict to Schwarzschild $r > 2M$ and static observers, for whom $-g_{tt} = 1 - 2M/r > 0$.

Theorem 5.3 (Tolman local temperature from the horizon scale). *Under Literature Hypothesis 5.1, Assumption 5.2, and Theorem 2.3 with asymptotic normalization $T_\infty = T_H$,*

$$T_{\text{loc}}(r) = \frac{T_H}{\sqrt{1 - 2M/r}} \quad (r > 2M). \quad (10)$$

Proof. Tolman’s law gives $T_{\text{loc}}(r) \sqrt{1 - 2M/r} = T_\infty$. The Hartle–Hawking / horizon-KMS state that underlies Theorem 4.5 has $T_\infty = T_H$. $\square$

Theorem 5.4 (Effective-method agreement with Tolman). *Once the HRT/Hawking package fixes the single static-Killing temperature scale T_H at infinity, Tolman’s equilibrium law continues that scale to every static radius $r > 2M$ by (10). The HRT horizon response therefore agrees analytically with Tolman’s local-temperature prediction for static observers.*

Proof. Immediate from Theorems 4.5 and 5.3. □

Remark 5.5 (Unruh acceleration versus Tolman). The proper acceleration of a static Schwarzschild observer is $a(r) = (M/r^2)/\sqrt{1 - 2M/r}$. The naive Unruh assignment $a/(2\pi)$ equals (10) only in the near-horizon Rindler limit. Away from the horizon the equilibrium temperature of the static KMS state is the Tolman profile, not $a/(2\pi)$ at every r [8, 6].

Proposition 5.6 (No static Tolman temperature for $r < 2M$). *For $r < 2M$ the Killing field ∂_t is spacelike, so there are no static observers and (10) does not apply. Interior radiative behaviour (partner modes, free-fall regularity) is outside the static Tolman claim; free-fall regularity near the horizon is the standard semiclassical expectation [8].*

6 Why Hawking and Tolman touched the Hysterical Response

This section is interpretive: it explains the contact, and is not Lean-certified.

Hawking. Hawking analyzed quantum fields against a causal horizon that absorbs and transmits, with vacuum regularity fixing a thermal scale [2]. That arena—a dissipative boundary, a retarded response, and a fluctuation partner required by detailed balance—is exactly the open-system response setting in which HRT is defined [1, 3]. He computed the asymptotic occupation of the fluctuation branch of that boundary response. HRT names the same structure as horizon Hysterical memory plus noise (Definition 3.4); the Euclidean period is the shared geometric input (Remark 2.4).

Tolman. Tolman asked for thermal equilibrium in a static gravitational field [6, 7]. Given any theory that supplies a single static-Killing temperature T_H at infinity, his law is the unique static radial continuation. He therefore touched the spatial organization of the same equilibrium response whose fluctuation scale Hawking (and the HRT horizon package) fix at the boundary—without needing the membrane Langevin vocabulary.

Joint reading. Hawking fixed the fluctuation scale at the causal dissipative boundary; Tolman fixed the static radial profile of that equilibrium. HRT’s horizon package is the open-system rephrasing that joins both, which is why their results agree analytically with HRT predictions (Theorems 4.5, 5.4).

7 Backreaction as feedback

Modeling Assumption 7.1 (Semiclassical mass loss). To leading scaling order, $P_H \propto A_H T_H^4$ with $A_H \propto M^2$ and $T_H \propto M^{-1}$, so

$$\dot{M} = -\frac{C}{M^2}, \quad C > 0. \tag{11}$$

Theorem 7.2 (Integrated evaporation law). *The solution of (11) with $M(0) = M_0$ satisfies $M(t)^3 = M_0^3 - 3Ct$ on the interval where $M > 0$, and reaches $M = 0$ at $t = M_0^3/(3C)$ when $C \neq 0$.*

Proof. Separate variables: $M^2 dM = -C dt$, integrate, and evaluate the extinction time. $\square$

Proposition 7.3 (Hotter and more luminous at smaller mass). *If $\theta, C, M_{\text{small}}, M_{\text{large}} > 0$ and $M_{\text{small}} < M_{\text{large}}$, then $\theta/M_{\text{large}} < \theta/M_{\text{small}}$. If also $M_{\text{small}}^2 < M_{\text{large}}^2$, then $C/M_{\text{large}}^2 < C/M_{\text{small}}^2$.*

Proof. Clear positive denominators. $\square$

Thus mass loss raises T_H and the radiated power, producing the standard positive-feedback evaporation runaway. The semiclassical model does not determine the ultimate endpoint.

8 Logical status of the results

Quantity / claim	Status
HRT resolvent response (1)	Prior [1].
Membrane / open-system embedding	Literature Hypothesis 3.1.
Euclidean period $\beta_H = 2\pi/\kappa$	Literature Hypothesis 2.2.
Quantum FDR / KMS noise	Literature Hypothesis 4.1.
Horizon Langevin packaging	Modeling Assumption 3.2.
$T_H = \kappa/(2\pi)$	Theorem 2.3 (Lean).
Bose occupation from detailed balance	Theorem 4.4 (Lean algebra).
Effective-method agreement with Hawking	Theorem 4.5 (dictionary + lit. hypotheses).
Tolman law $T\sqrt{-g_{tt}} = \text{const}$	Literature Hypothesis 5.1.
$T_{\text{loc}}(r)$ for $r > 2M$	Theorem 5.3 (Lean algebra).
Effective-method agreement with Tolman	Theorem 5.4.
Why Hawking/Tolman touched HRT	Interpretive (Section 6); not Lean.
Mass-loss cubic law	Theorem 7.2 (Lean).
Geometry-free KMS from Hysterical action alone	Not claimed.
Full Bogoliubov operators as Lean QFT	Not claimed.
Firewall / Page-curve resolution	Not claimed.

9 Conclusion

Under the membrane / open-system embedding of a nonextremal horizon, Hysterical Response Theory agrees analytically with Hawking radiation and Tolman local temperature: the same T_H , the same Planck factor and greybodies, and the same static profile $T_{\text{loc}}(r) = T_H/\sqrt{1 - 2M/r}$ for $r > 2M$. At the effective-response abstraction that agreement is method equivalence under shared geometric and thermal inputs, not a verbal analogy.

Hawking computed the fluctuation branch of a causal dissipative boundary response; Tolman computed the static equilibrium continuation of that thermal scale. HRT is the open-system packaging that joins both. The Euclidean period remains a geometric input—as it is for Hawking—so geometry-free thermality from a Hysterical action alone is left open and is not required for the agreement proved here.

References

- [1] Andrew Bruce Boyd. Hysterical response in open quantum systems. *Working mathematical formulation*, 2026. Companion response-theory paper in this series.
- [2] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220, 1975.
- [3] B. L. Hu, A. Raval, and S. Sinha. Black hole fluctuations and backreaction in stochastic gravity. 2002. Influence-functional / open-system treatment; CTP noise linked to Bogoliubov coefficients.
- [4] M. K. Parikh. Membrane horizons: The black hole’s new clothes. 1999.
- [5] K. S. Thorne, R. H. Price, and D. A. Macdonald, editors. *Black Holes: The Membrane Paradigm*. Yale University Press, New Haven, 1986.
- [6] Richard C. Tolman. On the weight of heat and thermal equilibrium in general relativity. *Physical Review*, 35:904–924, 1930.
- [7] Richard C. Tolman and Paul Ehrenfest. Temperature equilibrium in a static gravitational field. *Physical Review*, 36:1791–1798, 1930.
- [8] W. G. Unruh. Notes on black-hole evaporation. *Physical Review D*, 14:870–892, 1976.